

Solution Requirements

Date	04 July 2026
Team ID	To Be Assigned
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	2 Marks

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	User Input	The system shall allow farmers to enter soil nutrients (N, P, K), temperature, humidity, pH, and rainfall values.	High
2	Crop Prediction	The system shall recommend the most suitable crop using a trained Machine Learning model.	High
3	External Interfaces	The Flask application shall communicate with the Machine Learning model and dataset.	High
4	Data Validation	The system shall validate user input before processing the prediction.	High
5	Result Display	The system shall display the recommended crop clearly on the results page.	High
6	Business Rules	Crop recommendations shall depend on soil nutrients and environmental conditions.	High
7	Data Management	User input data shall be processed securely without unauthorized access.	Medium
8	Future Enhancement	The system shall support additional crop datasets and prediction models in the future.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The application shall generate crop predictions quickly.	Prediction within 2 seconds
2	Scalability	The system shall support multiple users simultaneously.	Support 1000+ users/day
3	Security	User input shall be handled securely.	Secure data processing
4	Reliability	The application shall provide consistent prediction results.	99% availability

5	Usability	The interface shall be simple and user-friendly.	Easy navigation on desktop and mobile
6	Maintainability	The application shall support easy updates and maintenance.	Modular code structure